HEAVY METAL TESTED & CERTIFIED STANDARDS

Heavy Metal Tested & Certified

Team Primer

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Team Primer

This is the short version. It is written for people joining or supporting the Heavy Metal Tested and Certified program who do not yet know how it works, and it aims to get everyone onto the same page in about fifteen minutes of reading.

It is deliberately brief. It tells you what the program is, walks the full path a brand travels from first inquiry to a certified mark, and puts the analytes, statuses, numbers, and terms you will hear most often on a single page. It is not the rulebook. Every rule here is stated in summary; the binding detail, the decision rules, the testing schedule, the enforcement and appeals machinery, and how the limits themselves are set, lives in the full Program Manual, which this primer points you to at the end.

This primer is for internal use. It contains no brand-identifying information and none should be added to it. Read it first; keep the full manual on hand for when you need the exact rule.

Orientation

This section orients a reader who is new to the program. It states in plain terms what the program is, walks the full path a brand travels from first inquiry to a certified mark, and puts the numbers and terms you will meet most often on a single page. It is a way in, not a substitute for the parts that follow: every rule summarized here is specified in full in Part 2 through Part 4.

Start here: what the program is

The Heavy Metal Tested and Certified program does one narrow thing and does it in the open. It publishes a numeric limit for each heavy metal in each kind of product, and then it certifies individual products against those published numbers under continuous laboratory testing. The limit is published before anything is certified against it, so a brand can read the exact threshold it will be held to before it applies or spends a dollar on testing.

Three choices make the program different from the testing a team member may already have in mind. First, the number comes first: the program does not test products and then decide what counts as a pass, it publishes the pass line in advance and holds everyone to it. Second, it is continuous, not a snapshot: a product is certified across multiple production lots and then kept under surveillance for as long as it carries the mark, so the mark describes an ongoing state rather than one lucky test. Third, it tests eight metals as ten analytes, where most food certifications test four, because a limit is only as protective as the metals it actually looks for.

The program is careful about what the mark claims. It certifies that a product was tested at an accredited laboratory, at or below a published number, on a stated analyte panel, in the form it is actually sold in. It

does not claim the product is safe, and a brand may not use it to claim "heavy metal free." That restraint is deliberate and it is what keeps the mark defensible.

One structural point matters from the first day. The program runs on two separate properties. The certification program, its limits, and its governance live at heavymetalcertified.com and are run by the Institute of Contaminant Standards. The scientific evidence the limits draw on lives separately at the Heavy Metal Index, heavymetalindex.com, a public library of the peer-reviewed and regulatory literature. The program reads from the Index; the Index does not endorse the program or any brand. Keeping those two apart is what lets the program cite an independent evidence base rather than its own marketing, and holding that separation is part of everyone's job here. Part 1.4 and, in the internal edition, Part 6.2 explain why.

The certification journey

Here is the whole path a brand travels, end to end. Each phase is specified in full in the part named beside it.

Phase 0. Readiness (optional, before applying). A brand can read the published limit for its subcategory and test privately to see where it stands, before committing to the program. This readiness step is nonpublic: there is no mark, and no public record that the brand even looked. It exists so that testing is a diagnostic a brand can use to build a plan, not a trap that penalizes a brand for checking.

Phase 1. Application and license. A brand that decides to proceed signs a license agreement and becomes a Licensee. The license authorizes participation and, once a product is certified, use of the mark. It also commits the brand to the program's data, testing, and supplier-intelligence obligations.

Phase 2. Baseline (three lots). Every product entering the program is tested across a minimum of three distinct production lots before any status is assigned. A single passing lot certifies nothing. Each lot is tested for the full panel, with speciation where required, sampled as a composite drawn across the lot, and documented under chain of custody. A brand that already runs a mature accredited testing program may credit

up to two of the three lots with qualifying prior data and complete the baseline with one new lot. Part 2.3 and Part 4.3 specify this.

Phase 3. Status assignment. The three baseline results are read together, and the worst-performing lot sets the initial status. A product at or below every limit is Status A and may carry the mark. A product with only a controllable Tier 2 metal between 100 and 150 percent of limit is Status B, certified transitionally, and may carry the mark with a disclosed correction plan. A product that does worse, or that fails a Tier 1 metal, is Status C or is not yet eligible. A brand whose baseline does not reach A or B but whose product is legally compliant can still enter, confidentially, on the Confidential Remediation Track, and work toward the mark out of public view. Part 2.4 and Part 2.5 specify the statuses and tracks.

Phase 4. The mark, and ongoing surveillance. Once a product reaches Status A or B, it may carry the mark under the mark-usage rules, and it enters continuous surveillance. How often it is tested depends on its status and its risk tier: routine for Status A, elevated for Status B, and every lot with hold-and-release for a product on probation. Any change to ingredients, water, equipment, packaging, formulation, or process triggers an unscheduled reflex test. Part 2.6 covers the mark; Part 4 covers the schedule.

Phase 5. When a lot falls short. If a certified product fails a lot, the brand opens a corrective action, fixes the root cause, and re-tests, on a clock that tightens with severity. The brand also chooses how the correction is handled: confidentially by default, or, electively, published on its own brand page as a documented catch-and-fix. Neither route avoids the correction, and neither shields a released lot that is above its limit. A brand that conceals a known exceedance loses its certificate. Part 2.5 and Part 2.9 specify this.

Phase 6. The ratchet (over time). Published limits tighten as the certified market improves, on a capped, pre-announced schedule, and never loosen. A brand can see the next move coming through a published forward forecast and plan for it. Part 2.11 specifies the mechanism.

The path in one line: read the number, test privately if you like, apply, run a three-lot baseline, get a status, earn the mark, stay under surveillance, correct any miss, and improve as the limits ratchet down.

Phase	What happens	The gate	Specified in
0. Readiness	Brand tests privately against the published limit	None; nonpublic, no mark	Part 2.5
1. Application	Brand signs the license, becomes a Licensee	License agreement	Part 2
2. Baseline	Three distinct lots, full panel, chain of custody	Three-lot minimum	Part 2.3, 4.3
3. Status	Worst lot sets Status A, B, C, or remediation track	Tier rules, decision rules	Part 2.4, 2.5
4. Mark and surveillance	Mark on A or B; testing by status and risk tier	Mark-usage rules	Part 2.6, Part 4
5. Correction	Corrective action, re-test, disclosure choice	CAPA clock, no concealment	Part 2.5, 2.9
6. Ratchet	Limits tighten on a capped, forecast schedule	Two-cycle trigger, 20 percent cap	Part 2.11

Program at a glance

The panel. Eight priority metals, tested as ten analytes. Two of the eight are speciated to the toxic form; methylmercury is a conditional eleventh, speciated by reflex and bounded by the total-mercury limit.

Metal	Analyte tested	Tier	Transitional variance
Lead (Pb)	Total	1	None
Cadmium (Cd)	Total	1	None
Arsenic (As)	Inorganic (iAs), speciated	1	None
Mercury (Hg)	Total; methylmercury on trigger	1	None
Nickel (Ni)	Total	2	To 150 percent
Tin (Sn)	Total	2	To 150 percent
Aluminum (Al)	Total	2	To 150 percent

Metal	Analyte tested	Tier	Transitional variance
Chromium (Cr)	Total; hexavalent on trigger	2	To 150 percent, Tier 1 when speciated

Tier 1 metals have no safe threshold and admit no exceedance at any margin. Tier 2 metals are controllable and admit a transitional exceedance to 150 percent under conditions.

The five statuses. A status attaches to a product, not a brand, and is visible on the certificate.

Status	Meaning	Mark
A	Certified, full compliance	Usable
B	Certified, transitional (a Tier 2 metal to 150 percent, with a correction plan)	Usable, disclosed
C	Probation (a Tier 1 exceedance, or 150 to 200 percent of limit)	Suspended
D	Suspension (above 200 percent, or a hard-stop such as a regulatory or data-integrity failure)	Prohibited
E	Revocation (concealment or repeated failure)	Prohibited

The numbers you will meet most. A limit is met at 100 percent of the published value or below (percent-of-limit, PoL). Transitional certification runs to 150 percent, on Tier 2 metals only. Above 200 percent is suspension. Entry requires three lots. Corrective action opens within 10 business days at Status B, 5 at Status C, and 2 at Status D. Speciation is triggered when a total metal reaches 50 percent of its limit. The ratchet moves at most 20 percent per event, with a 12-to-18-month transition, and only after two consecutive annual reviews confirm the market has moved.

The terms you will meet most. An *action level* is the limit that triggers intervention; it is not a *safety threshold*, and the difference is load-bearing. *ALARA* is the obligation to keep reducing even while

compliant. *Native basis* means a limit is stated in the form the product is sold in, powder as powder, ready-to-feed as consumed. *PoL* and *MaxPoL* are a result as a percentage of its limit, and the highest such across a product's analytes, which sets the status. *Reflex testing* is an unscheduled test triggered by a change or a warning result. *CAPA* is the corrective-and-preventive-action process. *Chain of custody* is the unbroken documented handling of a sample; a break invalidates the result. The *ratchet* is the mechanism that tightens limits over time. The full glossary is in Part 5.5.

Where to Go for More

This primer covers the shape of the program. When you need the exact rule, the full Program Manual is the reference, and it is organized so you can go straight to what you need.

- **How a product gets certified, in binding detail.** Program Manual Part 2: the principles, the analyte panel and tiers, the three-lot baseline, the five statuses, the remediation tracks, the mark rules, the decision rules, and corrective action.
- **The testing and surveillance schedule.** Part 4: risk tiers, how often each status is tested, the reflex triggers, speciation, sampling, and chain of custody.
- **How the limits themselves are set.** Part 3.1: the government-maximum default and the disclosed occurrence calibration beneath it, the publication gates, and the rationale tags.
- **Why the program holds up under challenge.** Part 5: the methodology defense, the program-level adversarial read, the recognition landscape, and the governance structure.
- **How the program is organized and run.** The internal edition of the manual carries Part 6, Program Operations: the entity structure, the firewall, the standards pipeline, the roles, and the governance workflow.

If you read one thing after this primer, read Part 2. It is the operational constitution, and it answers almost every “what happens if” question a brand or a colleague will ask you.